An Empirical Review of the Conference Review Process

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The Conference Paper Process

- Various other fields publish by submitting papers to journals
- However, in ML, conferences usually more selective/prestigious than journals
  - Paper reaches greater audience in a shorter time
  - ML is a fast-moving field
- Each paper assigned a group of reviewers, area chair receives reviews and makes a final decision

Past Research

- 2014 NIPS Experiment
  - Same set of papers reviewed by 2 different sets of reviewers
  - 57% of the papers accepted by the first committee were rejected by the second one and vice versa. Which may seem like a coin flip, but the acceptance rate of NIPS is 24.8%. So this is a relatively good sign.
- People in the field, however, have continued to question the validity and quality of the process. The goal of this research was to quantify these feelings.

Why Now?

Recently, computer science conferences are starting to adopt OpenReview

OpenReview

- A publicly accessible database for conference papers submissions, reviews, and comments.

With OpenReview, we can scrape what papers were accepted/rejected and the other information like authors, emails, and ratings!

Paper Example

Large Batch Optimization for Deep Learning: Training BERT in 76 minutes

Yang You, Jing Li, Sashank Reddi, Jonathan Hseu, Sanjiv Kumar, Srinadh Bhojanapalli, Xiaodan Song, James Demmel, Kurt Keutzer, Cho-Jui Hsieh

25 Sep 2019 (modified: 11 Mar 2020) ICLR 2020 Conference Blind Submission Readers: Everyone Show Bibtex Show Revisions

Original Pdf: pdf

TL;DR: A fast optimizer for general applications and large-batch training

Abstract: Training large deep neural networks on massive datasets is computationally very challenging. There has been recent surge in interest in using large batch stochastic optimization methods to tackle this issue. The most prominent algorithm in this line of research is LARS, which by employing layerwise adaptive learning rates trains ResNet on ImageNet in a few minutes. However, LARS performs poorly for attention models like BERT, indicating that its performance gains are not consistent across tasks. In this paper, we first study a principled layerwise adaptation strategy to accelerate training of deep neural networks using large mini-batches. Using this strategy, we develop a new layerwise adaptive large batch optimization technique called LAMB; we then provide convergence analysis of LAMB as well as LARS, showing convergence to a stationary point in general nonconvex settings. Our empirical results demonstrate the superior performance of LAMB across various tasks such as BERT and ResNet-50 training with very little hyperparameter tuning. In particular, for BERT training, our optimizer enables use of very large batch sizes of 32868 without any degradation of performance. By increasing the batch size to the memory limit of a TPUv3 Pod, BERT training time can be reduced from 3 days to just 76 minutes.

Keywords: large-batch optimization, distributed training, fast optimizer

Code: https://github.com/tensorflow/addons/blob/master/tensorflow_addons/optimizers/lamb.py

7 Replies Add **Public Comment**

Show all from everybody

[−] **Paper Decision**

ICLR 2020 Conference Program Chairs

19 Dec 2019 (modified: 19 Dec 2019) ICLR 2020 Conference Paper1 Decision Readers: Everyone

Decision: Accept (Poster)

Comment: This paper presents a range of methods for over-coming the challenges of large-batch training with transformer models. While one reviewer still questions the utility of training with such large numbers of devices, there is certainly a segment of the community that focuses on large-batch training, and the ideas in this paper will hopefully find a range of uses.

Add **Public Comment**

Review Example

Official Blind Review #2

ICLR 2020 Conference Paper1 AnonReviewer2

23 Oct 2019 (modified: 05 Nov 2019) ICLR 2020 Conference Paper1 Official Review Readers: Everyone

Review: This paper proposes a learning rate adaptation mechanism, called LAMB, for large-batch distributed training. The goal is to stabilize the training as the batch size increases. The idea is simple and straightforward -- there should be a layerwise learning rate adjusted by normalizing the layer weights and gradients at each layer so that layers with larger weights take larger learning steps, and vice versa. The authors perform empirical studies on BERT-large and ResNet to conclude that LAMB can scale up training batch size while still being able to converge in time with comparable accuracy.

Strengths:
+ Demonstrate the scalability of large-batch training (up to 64K) on BERT-Large with comparable accuracy.
+ A leap from the prior work LARS that demonstrates the layer-wise learning rate adjustment scheme also works with Adam for NLP tasks.
+ The re-warmup technique for stabilizing the second phase of mixed sequence training is neat.

Weaknesses:
- Although the authors' analysis is based on a large set of models and clearly outperforms the prior work LARS, it is still hard to assess the generality of the obtained results. The authors made an effort to show evaluation results on MNIST and CIFAR-10, but they are much less challenging tasks.
- Technical novelty over LARS seems to be incremental, where a large portion of the work is essentially applying LARS to Adam and demonstrate its effectiveness on BERT and ResNet.

Overall, the LAMB technique seems to be simple to apply yet very useful in practice for large scale training. The work can potentially help the practitioner to scale-out large model training to hundreds or even thousands of GPUs/TPUs with good scalability. Moving forward, the authors are encouraged to report LAMB optimization results on transformer-based models such as GPT, RoBERTa, and XLNet.

Question:
Does the training take advantage of FP16 half-precision training?
How does the training process handle overflow and NaN gradients?
What is the significance of the range [α_l, α_u] in Theorem 2 and Theorem 3, and how to choose the value for them in practice?

Rating: 6: Weak Accept

Experience Assessment: I have read many papers in this area.

Review Assessment: Thoroughness In Paper Reading: I read the paper at least twice and used my best judgement in assessing the paper.

Review Assessment: Checking Correctness Of Derivations And Theory: I did not assess the derivations or theory.

Review Assessment: Checking Correctness Of Experiments: I assessed the sensibility of the experiments.

Add **Public**

Questions

1. Reproducibility: How likely is it for a paper to be accepted again?
2. Quality: Are Papers that are high-impact accepted?
3. Bias: Are institutions of better rank unfairly accepted or rejected?
4. Topic Bias: Are certain topics with certain keywords accepted at a higher rate than others?

What did we do

Phases of research

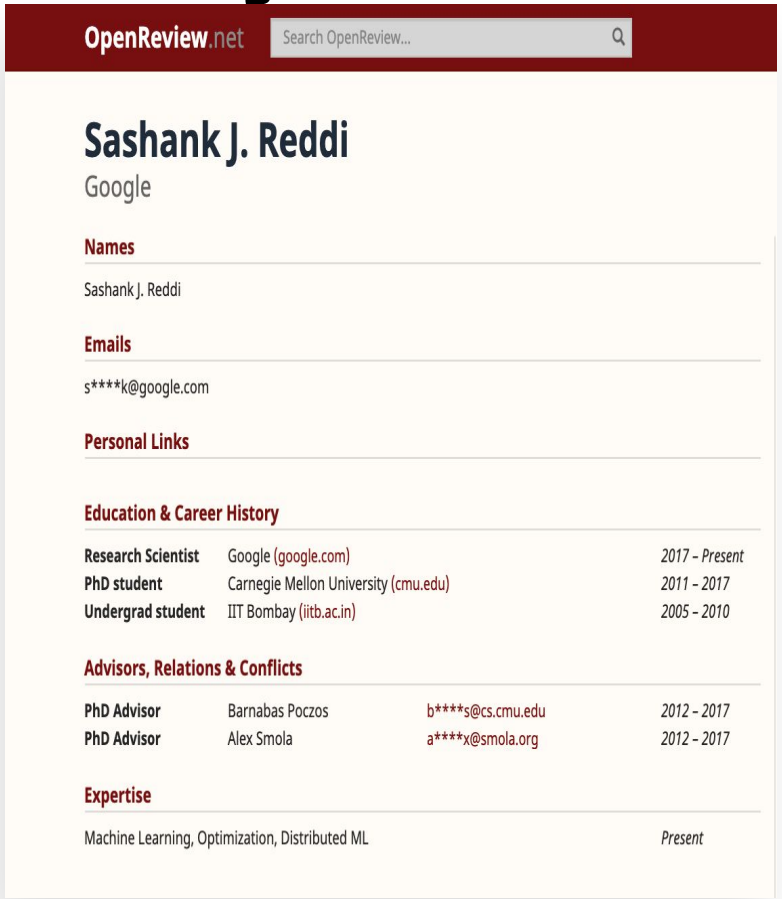
- Scraping(Most of the time)
- Statistics and conclusions(Briefly mention)

Scraping phase 1: Openreview.net

- Method:
  - Openreview API
  - Get all data from 6 separate databases and merges them together.
    - Paper, Decision, Review, Comment, Author Profiles, Withdrawn Paper.
- Issues with getting the data:
  - Different years have different keys to access the database.
    - 2018 decision database: ICLR.cc/2018/Conference/-/Acceptance_Decision
    - 2019 decision database: ICLR.cc/2019/Conference/-/Paper.*/Meta_Review
  - The keys were not documented on the website.
- Solutions:
  - Reach out to the API maintainer and ask for clarifications.

Scraping Phase 1: Institution Matching

- First approach: matched authors to institutions using only Profile database.
  - Only 40% of authors have an institutional matching (out of 12,000).
  - Variations institution name
    - e.g. University of Maryland, College Park, University of Maryland - College Park, and UMD means the same thing.
  - 1541 unique institutions
  - Author profiles are not updated
  - Authors with multiple institutions at the same time.



OpenReview.net Search OpenReview...

Sashank J. Reddi
Google

Names

Sashank J. Reddi

Emails

s****k@google.com

Personal Links

Education & Career History

Research Scientist	Google (google.com)	2017 – Present
PhD student	Carnegie Mellon University (cmu.edu)	2011 – 2017
Undergrad student	IIT Bombay (iitb.ac.in)	2005 – 2010

Advisors, Relations & Conflicts

PhD Advisor	Barnabas Poczos	b****s@cs.cmu.edu	2012 – 2017
PhD Advisor	Alex Smola	a****x@smola.org	2012 – 2017

Expertise

Machine Learning, Optimization, Distributed ML — Present

Scraping Phase 1: Institution Matching (cont)

- Second approach built a hash table that maps domain to institution
- Open source World University Domain dataset
  - Transform “cs.umd.edu", “umiacs.umd.edu", “ece.umd.edu", “math.umd.edu",... to “umd.edu"
  - Transform 'Department of Computer Science, University of Maryland', ‘UMD’, ... to “ University of Maryland, College Park"
- 95% of authors have an institutional matching
- 1541 unique institutions are reduced to only 841.

Education & Career History

Research Scientist	DeepMind (google.com)
Postdoc	University of Oxford (ox.ac.uk)
PhD student	University of Oxford (ox.ac.uk)

About

University Domains and Names Data List & API

Readme

MIT License

Scraping Phase 1: Ranking Matching

- Levenshtein edit distance to compare institutions in openreview dataset to ranking dataset
- Manual checking
  - For institutions with no matched ranking
  - Unranked institutions are filled with lowest value (1 is highest)

timeshighereducation.com

3	Stanford University United States Explore	94.7	93.6	96.8	99.9	64.6	79.3
4	Massachusetts Institute of Technology United States Explore	94.2	91.9	92.7	99.9	87.6	89.0

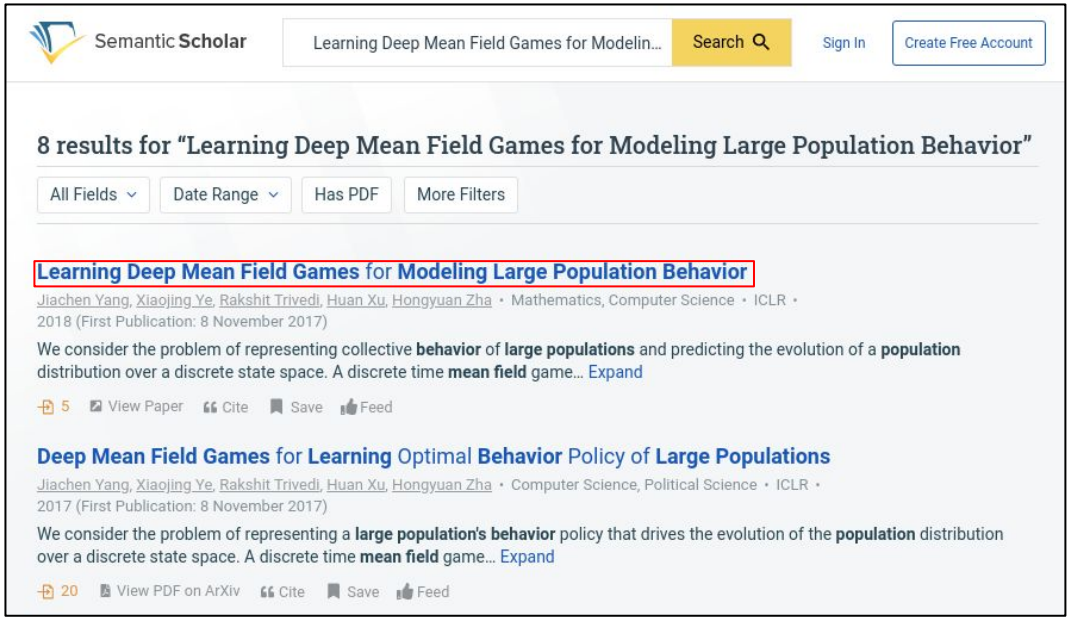
csrankings.org

#	Institution	Count	Faculty
1	Carnegie Mellon University	16.7	156
2	Massachusetts Institute of Technology	12.5	89
3	Univ. of Illinois at Urbana-Champaign	11.3	99
4	Stanford University	10.9	67
5	University of California - Berkeley	10.3	92

Scraping Phase 2: Semantic Scholar Links

- Search for paper from URL using the title
- Selenium: render javascript page
- Compute the minimum Levenshtein edit distance
  - Insertion, Deletion, Substitution
- Compare all results with the min
  - Title, Authors, Abstract
- Record the appropriate paper and author links, first publication

https://www.semanticscholar.org/search?q=Learning%20Deep%20Mean%20Field%20Games%20for%20Modeling%20Large%20Population%20Behavior&sort=relevance



Scraping Phase 2: cont.

- Determine other versions of the same paper
  - Compare distances with min
- Heuristic details:
  - Tight threshold values for title, authors, and abstract
  - 2/3: Append to csv to be checked manually
  - Author list sorted alphabetically to allow different permutations
- Take union of author links
  - Accounts for both duplicate and missing authors

Learning Deep Mean Field Games for Modeling Large Population Behavior

Jiachen Yang, Xiaojing Ye, Rakshit Trivedi, Huan Xu, Hongyuan Zha • Mathematics, Computer Science • ICLR • 2018 (First Publication: 8 November 2017)

We consider the problem of representing collective **behavior** of **large populations** and predicting the evolution of a **population** distribution over a discrete state space. A discrete time **mean field** game (MFG) is motivated as an interpretable **model** founded on game theory for understanding the aggregate effect of individual actions and predicting the temporal evolution of **population** distributions. We achieve a synthesis of MFG and Markov decision processes (MDP) by showing that a special MFG is reducible to an MDP. This enables us to broaden the scope of **mean field** game theory and infer MFG **models** of **large** real-world systems via **deep** inverse reinforcement **learning**. Our method learns both the reward function and forward dynamics of an MFG from real data, and we report the first empirical test of a **mean field** game **model** of a real-world social media **population**. Collapse

5 View Paper Cite Save Feed

Deep Mean Field Games for Learning Optimal Behavior Policy of Large Populations

Jiachen Yang, Xiaojing Ye, Rakshit Trivedi, Huan Xu, Hongyuan Zha • Computer Science, Political Science • ICLR • 2017 (First Publication: 8 November 2017)

We consider the problem of representing a **large population's behavior** policy that drives the evolution of the **population** distribution over a discrete state space. A discrete time **mean field** game... Expand

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ING LARGE POPULATION BEHAVIOR

Jiachen Yang, Xiaojing Ye, Rakshit Trivedi, Huan Xu, Hongyuan Zha • 2018

We consider the problem of representing collective **behavior** of **large populations** and predicting the evolution of a **population** distribution over a discrete state space. A discrete time **mean field** game... Expand

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On the Convergence of Model Free Learning in Mean Field Games

Romuald Elie, Julien Pérolat, Mathieu Laurière, Matthieu Geist, Olivier Pietquin • Mathematics, Computer Science • AAAI • 2020 (First Publication: 4 July 2019)

Learning by experience in Multi-Agent Systems (MAS) is a difficult and exciting task, due to the lack of stationarity of the environment, whose dynamics evolves as the **population** learns. In order to... Expand

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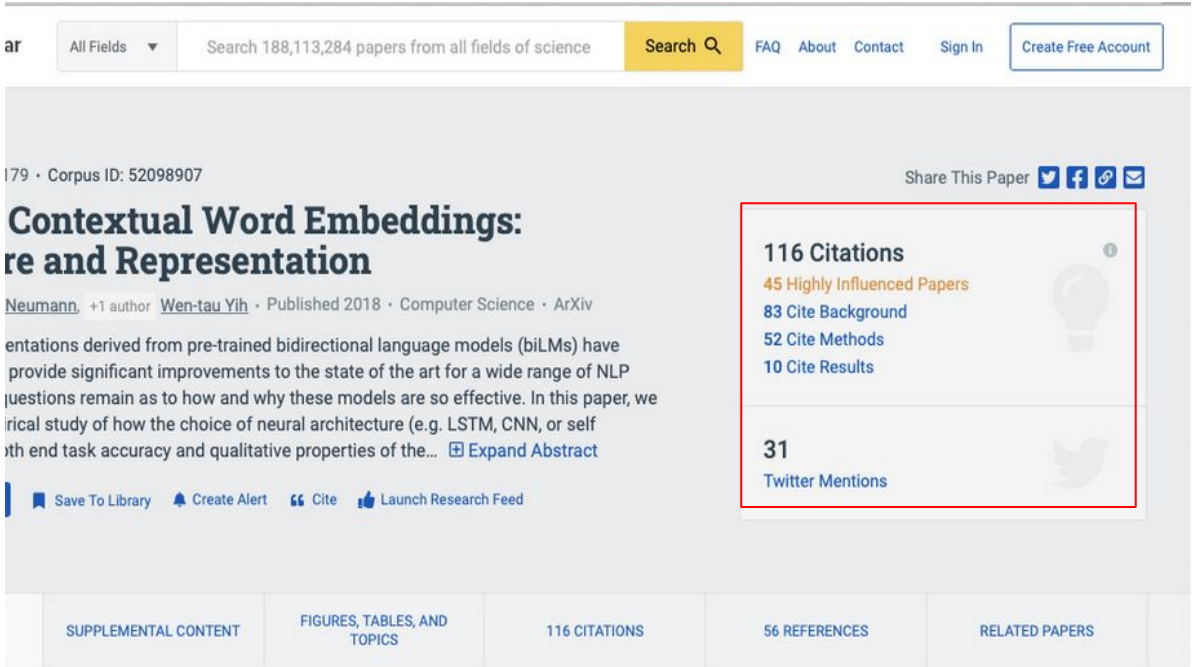
Kinetic Theory and its neighbours

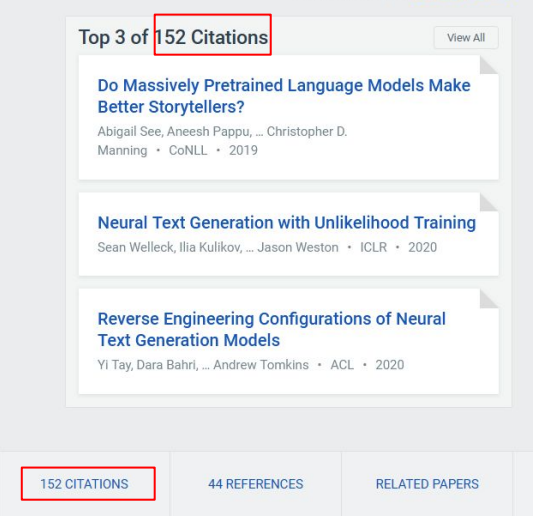
Scraping Phase 3: Semantic Scholar

- Selenium
- Paper links: citations, highly influential citations, Twitter mentions
- Author links: # publications, citations/highly influential citations, *h*-index (largest *h* s.t. author has at least *h* papers each with *h* citations)
- For each paper, store paper statistics plus one set of stats per author

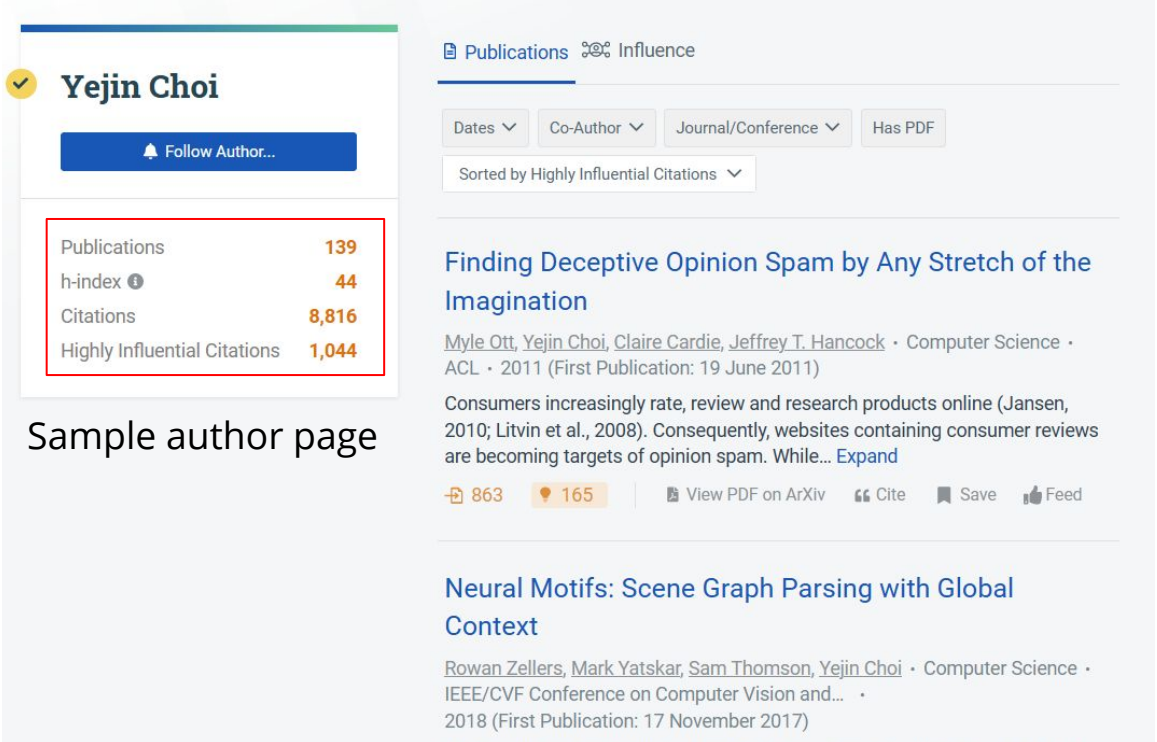


Sample paper page with citation “scorecard”

Scraping Phase 3: cont.



Updated citation info box



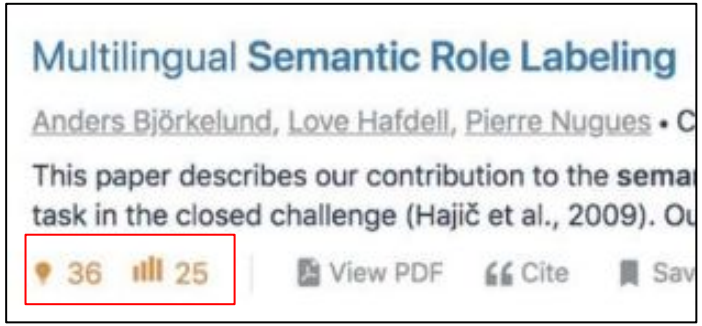
Sample author page

Issues with scraping

- Initially planned to use Google Scholar, but very unfriendly to web scraping - unrealistic amount of time required to avoid IP bans
  - Random queries every 8-11 minutes... 36 days!

How we overcame issues

- Semantic Scholar API: impractical
  - Instead, used searching and links to cover all (most) papers
- Semantic Scholar citations:
  - Scraping on the results page was inconsistent

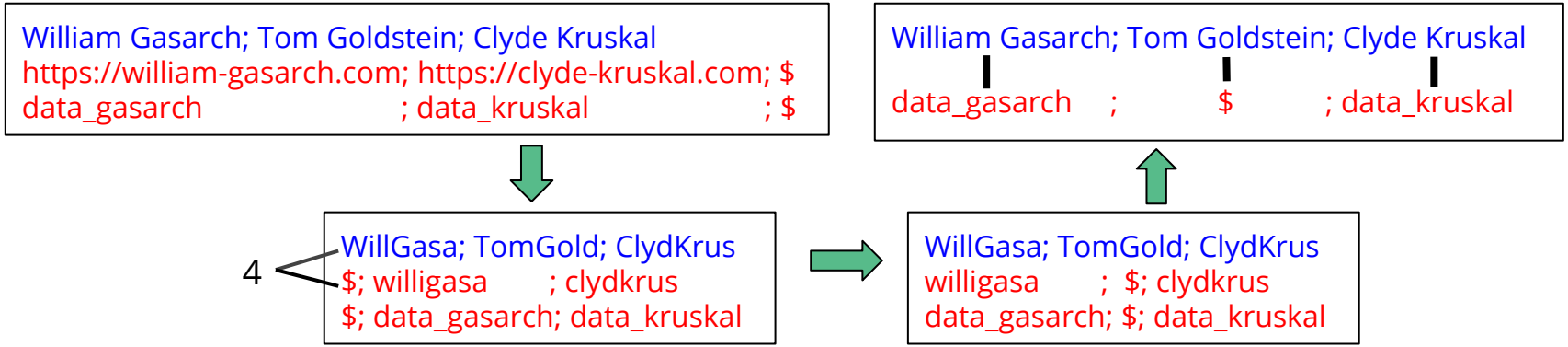


Data curation

- Verify papers using edit distance
  - Script outputs warnings for papers above threshold: manual check
- Compare Openreview authors to Semantic Scholar authors
  - Number of authors and the names themselves
    - Ensure that len(openreview) >= len(semanticScholar): otherwise, manual check
    - For authors with multiple pages, sum publication and citation counts, take maximum of *h*-index
- Check the earliest appearance date

Data curation: cont.

- Mapping heuristic:
  - Take only first and last name, with at most 4 characters from each name.
  - Introduce wildcard character for missing authors
    - Distance 0 for final distance, but distance 4 for algorithm (1)
  - (1) Greedy matching algorithm: match with the closest name remaining
  - (2) Sort alphabetically
  - Final order: min of (1) and (2)
  - ~99% accuracy!



Final Data Set

conference	year	paper	authors	emails	ratings	confidence	decisions	cmt_before_review	cmt_between	cmt_after_decision
ICLR	2017	Genera	Dougal J.	dougal@gma	6;8;7	3;3;4	Accept (Poster)	2	3	1
ICLR	2017	A Simp	Sanjeev A	arora@cs.pri	7;7;8	4;4;3	Accept (Poster)	8	4	0
ICLR	2017	A Base	Dan Hendr	dan@ttic.edu	6;6;6	3;3;3	Accept (Poster)	2	4	0

double_blinded	submission_date	institution	csranking	ranking	categories	earliest_appearance_date
no	11/4/16	University Co	45;1;45;-1	15;23;1	5;4	11/4/16
no	11/4/16	Princeton Uni	32;32;32	7;7;7	6;1;5	11/4/16
no	11/4/16	Toyota Techn	-1;-1	-1;-1	3;2	10/7/16

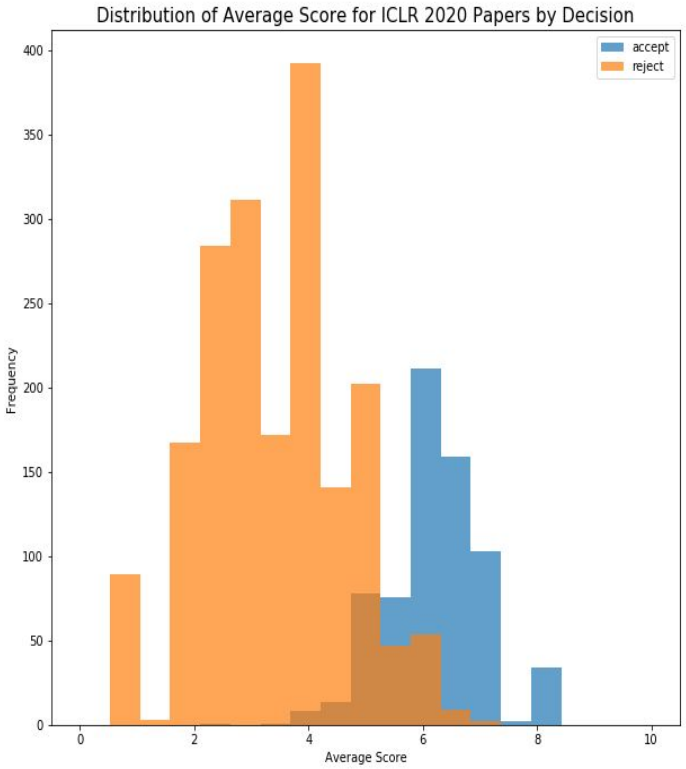
citations	cite_background	cite_methods	cite_results	twitter_mentions	highly_influenced_papers
109	44	52	1	0	12
582	241	295	21	0	112
387	218	194	14	0	98

authors_citations	authors_publications	authors_h-index	authors_highly_influenced_papers
734;445;739;104;1	36;25;25;2;104;389;21	14;8;13;1;19;99;48	100;44;139;12;135;9073;2223
16100;2835;3902	349;81;87	61;25;32	1798;422;503
1493;5710	27;99	14;31	280;821

Question 1: Reproducibility

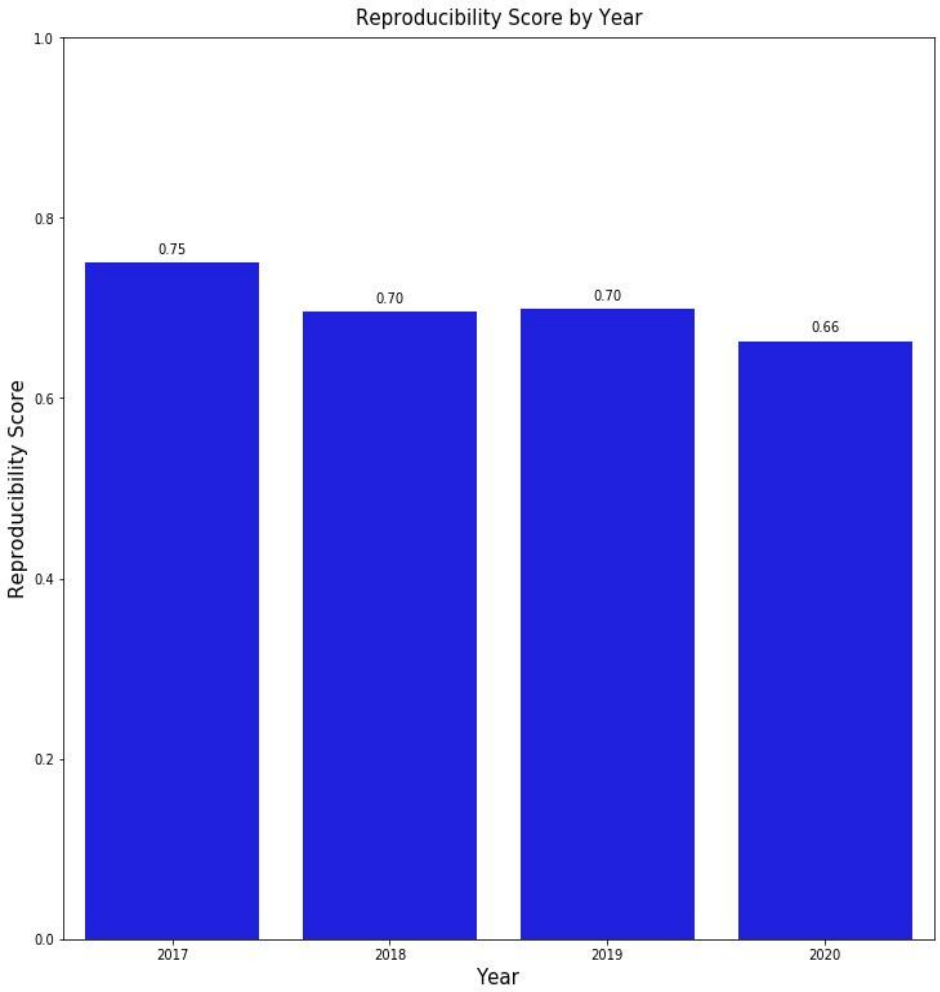
What's the probability of a paper being accepted again if submitted the second time ?

- **Method:** Bootstrap method for ICLR 2020 with 2560 papers
- **Step 1:** Draws 2560 papers from the empirical distribution of means for the conference.
- **Step 2:** For each paper, synthesizes 3 review scores by adding deviations drawn from the empirical distribution of deviations for papers with similar mean scores.
- **Step 3:** Synthesizes a new set of decisions.
    - Logistic Regression: given a paper mean score, returns the probability of a paper getting accepted.
    - Generates new decisions based on the above probability.
- Repeat step 2 to 3 again to obtain a second set of decisions.
- Reproducibility score = accepted twice / accepted the first time



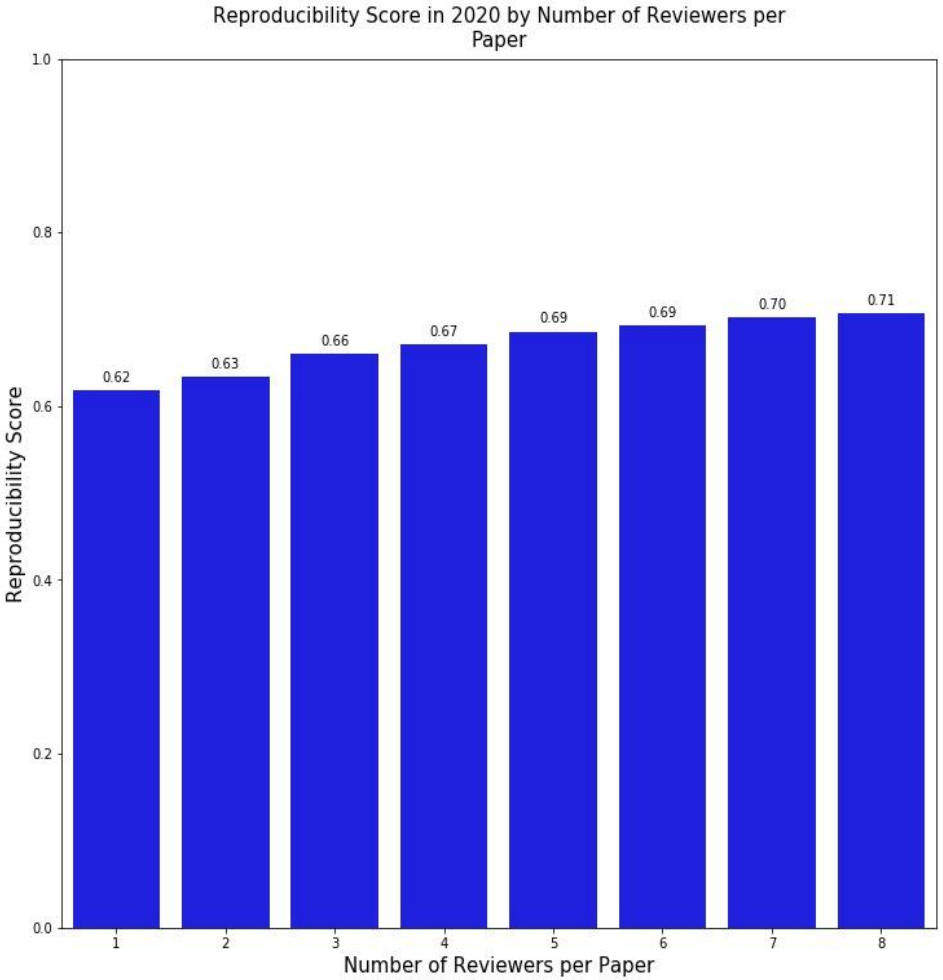
How has the reproducibility score changed over time ?

- The review process is becoming more random over time



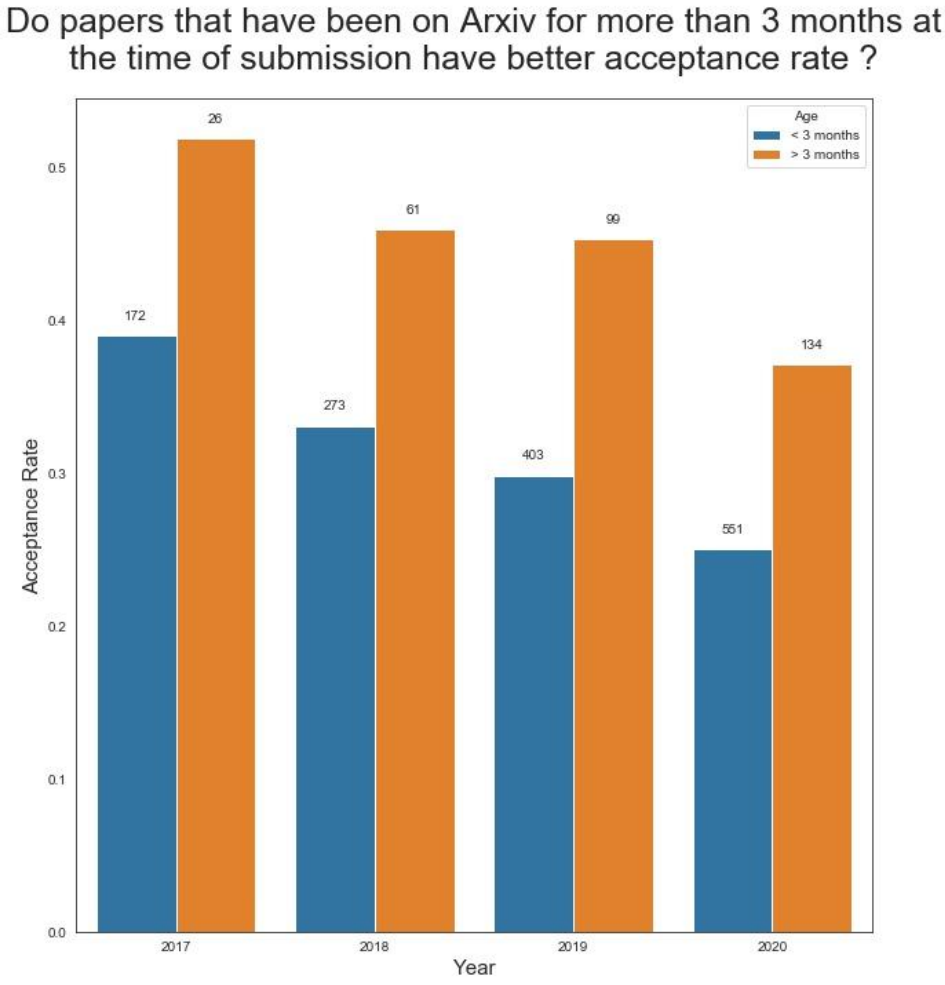
How does the reproducibility level change with the number of reviewers per paper ?

- Increasing the number of reviewers improves the reproducibility score but not by much.



Do papers that have been on Arxiv for more than 3 months at the time of submission date better or worse ?

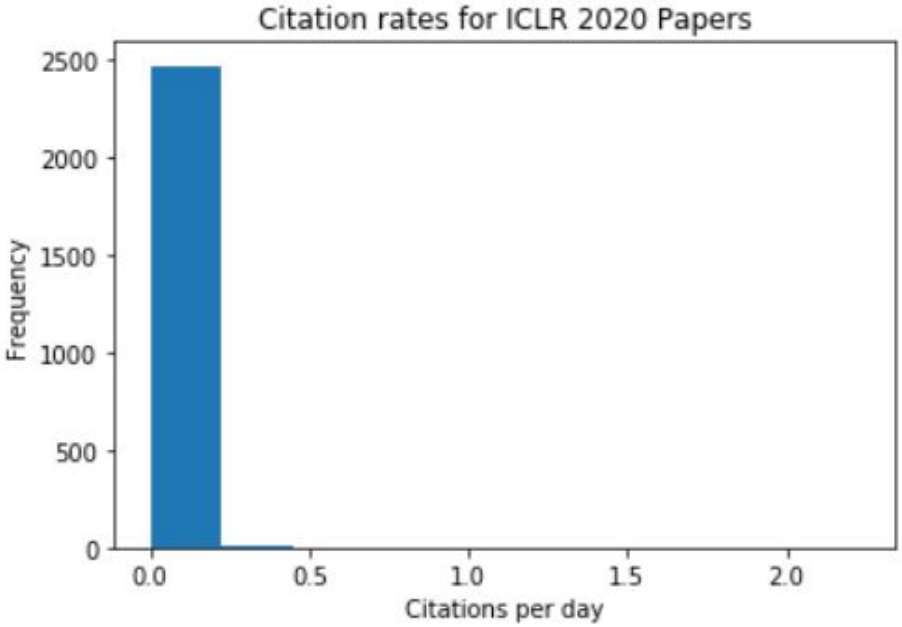
- Those papers are papers that are likely to be rejected from other conferences.
- All of them have higher acceptance rates compared to papers that are less than 3 months old.
- Papers should not be discouraged from resubmitting.



Question 2: Review Quality

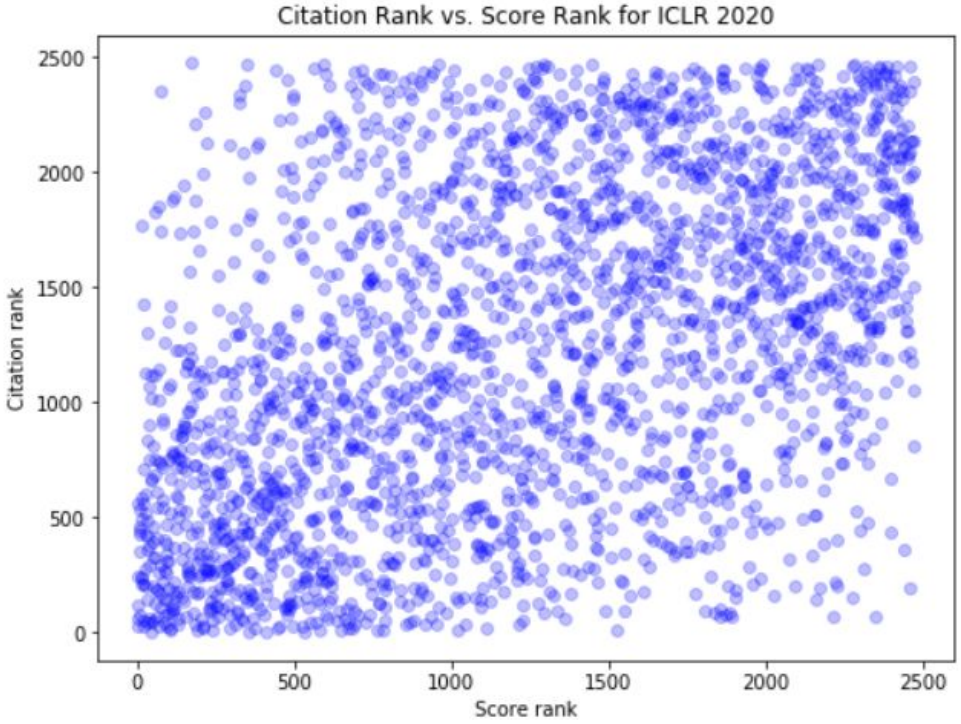
Are positive reviews correlated with high-impact papers?

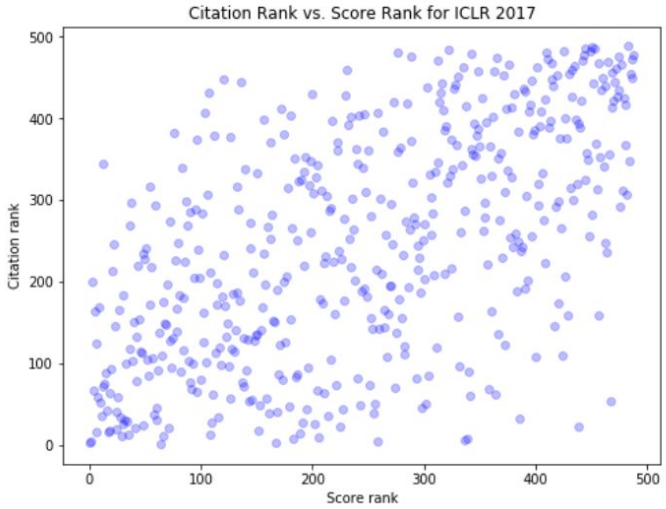
- Do papers with higher scores have higher citation rates?
- Citation rates are highly skewed - use ranks instead of absolute values, break ties arbitrarily. Do the same for score.
- Spearman's rank correlation coefficient

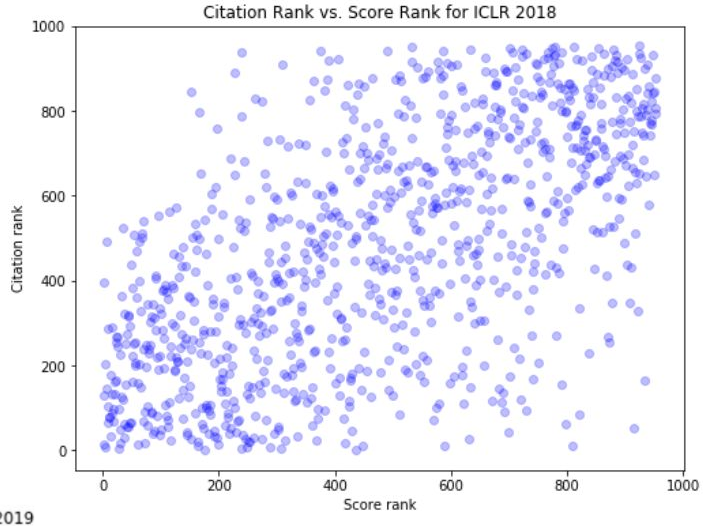


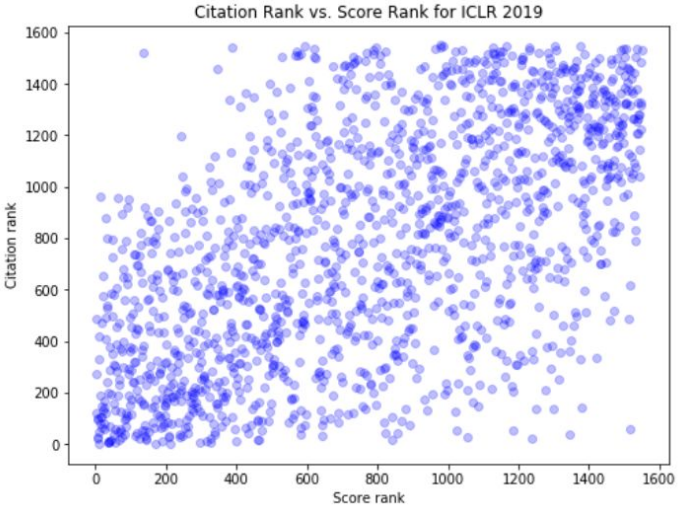
Spearman Correlation

- Positive correlation between citation rank and score rank appears to exist visually
- Spearman correlation coefficient = 0.471, $p < 0.001$





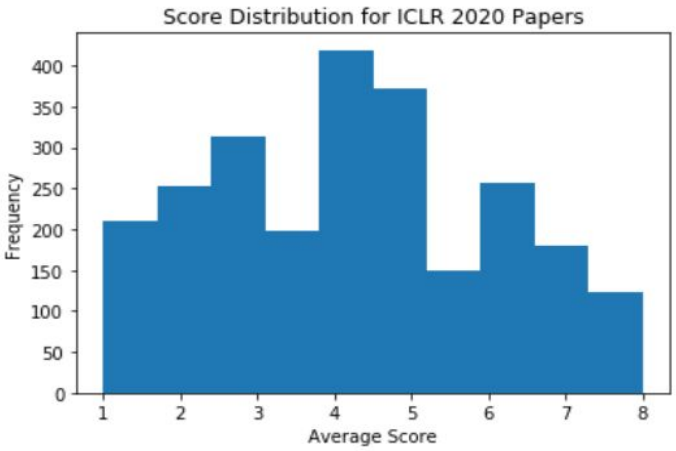




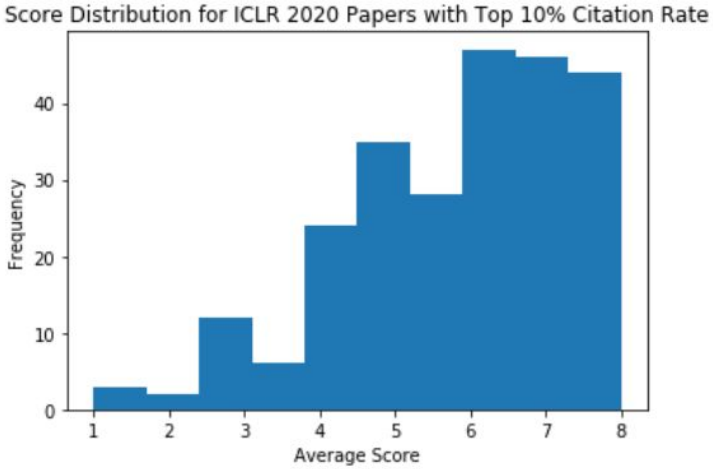
Spearman coeffs:
2017: 0.582
2018: 0.617
2019: 0.591

All p-values < 0.001

- Another approach: are the highest-impact papers getting high scores?
- Compare score distribution of all papers to papers with citation rate in the top 10%



Year	Mean avg. score (all)	Mean avg. score (top 10%)
2017	5.680	6.792
2018	5.370	6.421
2019	5.331	6.455
2020	4.250	5.741



Question 3: Institutional Bias

Question Raised: Are institutions of a better rank unfairly accepted or rejected?

Throughout this part of the presentation, a higher ranked (lower number) institutions will be referred to as more prestigious and lower ranked (higher number) institutions will be referred to as less prestigious

Types of Acceptances

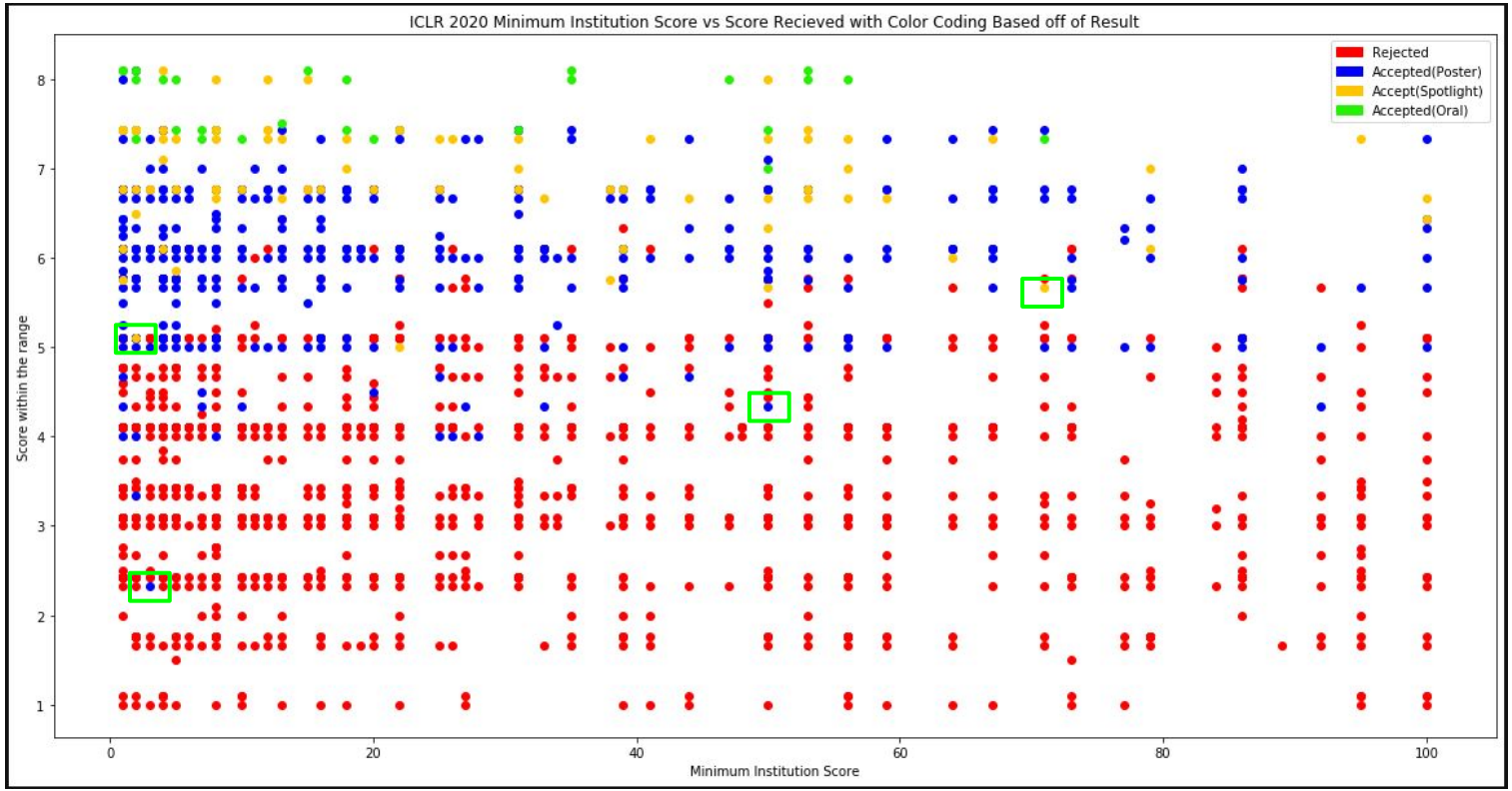
In order of least to most prestigious:

- Reject
- Withdrawn
- Invite to Workshop Track(Not included in 2020)
- Accepted(Poster)
- Accept(Spotlight)
- Accepted(Oral)

Filtering the Dataset and issues that arose

- No way to rank Companies and Rankingless institutions
- We represented them in the data set as -1
- Before calculating the most prestigious institution rank, we filtered out all - 1 values to get an accurate value
- For example
  - An original author list for an individual paper may originally look like [45,-1,25,-1](A rating per author on the paper)
    - The most prestigious(lowest) ranking of this would be -1
  - Whereas, we filtered out the -1 to get a list, [45,25], and calculated the accurate most prestigious rank of 25

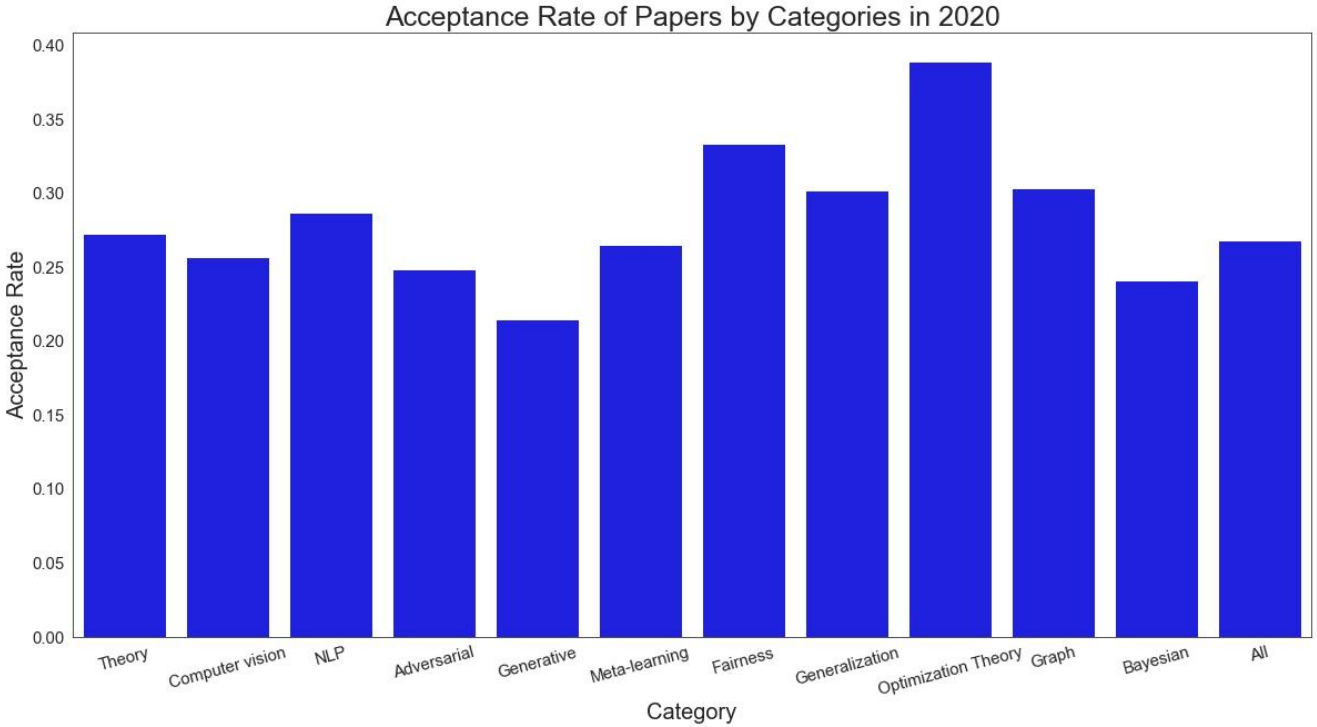
Graph 1


ICLR 2020 Minimum Institution Score vs Score Recieved with Color Coding Based off of Result
Rejected
Accepted(Poster)
Accept(Spotlight)
Accepted(Oral)
Score within the range
Minimum Institution Score
0
20
40
60
80
100
1
2
3
4
5
6
7
8

Question 4: Topic Bias

Question Raised: Are papers on certain topics and with certain keywords unfairly accepted?

Graph 1


Acceptance Rate of Papers by Categories in 2020
Acceptance Rate
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Theory
Computer vision
NLP
Adversarial
Generative
Meta-learning
Fairness
Generalization
Optimization Theory
Graph
Bayesian
All
Category

Plans Moving Forward

- Continue to run statistical tests/models
  - (other bias etc. to examine)?
- Draft a paper
- Submit to ICLR 2021

Tentative Takeaways and What we Learned

Tentative Takeaways

- Over time, reproducibility scores have dropped potentially alluding to the fact that the conference review process has gotten worse
- High-scoring reviews generally correspond with high-impact papers
- Initial evidence shows that higher ranked universities may require a lower score to be accepted into the conference
- Specific topics are accepted at a higher acceptance rate than others

What we Learned

- About the conference process
- Algorithm Design
- Using Web scraping tools like Selenium

Thanks

Thanks Dr. Goldstein, Micah, and UMD for having this program!